

Premium Student Guide: ARIMA, SARIMA & PyCaret

Comprehensive beginner-friendly notes for Data Science students.

ARIMA

ARIMA stands for AutoRegressive Integrated Moving Average. It is used for forecasting time-series data with trends.

Parameters: p , d , q

p = autoregressive terms, d = differencing, q = moving average terms.

Use Cases: sales forecasting, stock analysis, demand forecasting.

SARIMA

SARIMA extends ARIMA by adding seasonality handling.

Parameters: $(p,d,q)(P,D,Q,s)$

s represents the seasonal cycle such as 12 months.

SARIMA is useful for airline passengers, electricity demand, and retail sales.

ARIMA vs SARIMA

ARIMA handles trend. SARIMA handles trend plus seasonality.

PyCaret

PyCaret is a low-code AutoML library that automates machine learning workflows.

Workflow

setup() → compare_models() → create_model() → tune_model() → evaluate_model() →
finalize_model() → save_model()

compare_models() automatically compares many ML algorithms and ranks them.

Teaching Summary

ARIMA = Trend Forecasting. SARIMA = Trend + Seasonality Forecasting. PyCaret = Low-Code Machine Learning.